ROBERT DAKOUO

Data & Artificial Intelligence Engineer

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PROFILE

Final-year Data & Artificial Intelligence engineering student (ESMT / INGC-India, Dakar). I design end-to-end AI systems: machine learning, LLM agents with RAG, computer vision and noisy-data processing, from rigorously compared models to deployed APIs. Strong interest in AI for African languages and contexts, health and safety. Available immediately for an internship, work-study or freelance mission.

EDUCATION

Engineering Degree (Design Engineer), Data & Artificial Intelligence *2025 – 2026*
ESMT / INGC-India, Dakar, Senegal · degree expected in 2026

Bachelor's Degree in Engineering Science (Telecommunications option) *2023 – 2024*
ESMT, Dakar, Senegal

Bachelor's Degree in Electrical Engineering and Industrial Computing *2020 – 2023*
CERFITEX / GEII, Ségou, Mali · Python, control systems, embedded systems, networks, industrial computing

TECHNICAL SKILLS

Machine learning : Random Forest, XGBoost, CatBoost, SVM, K-means, K-medoids, Fuzzy C-means, co-training, SMOTE, RandomizedSearchCV

NLP & LLM : Hybrid RAG (BM25 + dense + reranking), QLoRA / LoRA fine-tuning, RAFT, multi-LLM agents, ChromaDB, mT5, MarianMT, Hugging Face, LangChain, Ollama

Computer vision : YOLOv8, Faster R-CNN, DETR, RT-DETR, ResNet, OpenCV, PaddleOCR, InsightFace, MediaPipe

Data, backend & systems : Python (Pandas, NumPy, scikit-learn, PyTorch, TensorFlow), SQL, FastAPI, REST APIs, React / Next.js, Kafka, Storm, TEE, AES-256-GCM

Tools : Git / GitHub, Linux, Jupyter, VS Code, Docker (basics)

PROJECTS

- **AI assistant on the 3GPP technical corpus.** 7 strategies compared across 3 LLMs (baseline, RAG, optimized RAG, QLoRA / LoRA fine-tuning, RAFT), a 3-agent system with arbitration and a FastAPI API. Best accuracy: 78% (Qwen2.5-7B), 15,000+ documents indexed (ChromaDB).
- **EWAATI: medical prescription extraction.** Fully local, privacy-by-design pipeline: photo, OpenCV, PaddleOCR, Mistral 7B, JSON. +200% extraction on 129 real prescriptions.
- **Bank fraud detection (Zindi).** Random Forest and XGBoost compared and tuned. Score of 0.81 on the leaderboard.
- **Object-detection robustness.** YOLOv8, Faster R-CNN, DETR and RT-DETR compared on Cityscapes under 5 weather perturbations. mAP 0.92 (YOLOv8), over 80% retention (Faster R-CNN).
- **SmartMask: mask-wearing detection.** Computer-vision identification of passers-by with or without a mask, with bounding box and confidence score.
- **Bambara to French translation.** mT5-small (general-purpose) vs fine-tuned MarianMT on a Bambara–French corpus. BLEU 12.31 vs 0.05.
- **SafeJambaar: emergency alert app (hackathon).** Real-time mobile alert app (theft, assault, harassment, stalking) with conversational AI, prototyped in 3 days at Orange Digital Center (Dakar 2026 Youth Olympic Games).
- **Online proctoring with facial 2FA.** Authentication via face embeddings (InsightFace) and live attention tracking (EAR, gaze, posture) with MediaPipe.
- **Secure medical diagnosis.** ResNet-18 for pneumonia detection (88.5% accuracy, 3.8 ms inference), secure-execution study (TEE) and AES-256-GCM encryption.
- **Real-time Big Data pipeline.** Binance API, Kafka, Storm and WebSocket to a Next.js interface.

COMPETITIONS & HACKATHONS

Zindi competition: bank fraud detection *2025 – 2026*
Random Forest / XGBoost models, score of 0.81.

Orange Digital Center hackathon (3 days) *2026*
SafeJambaar project, Dakar 2026 Youth Olympic Games context.

EXPERIENCE

Intern, Electrical Engineering & Industrial Computing *July 2023*
Regional Energy Directorate of Ségou (DRES), Mali

- Field data collection and monitoring of the Baraoueli hybrid power plant.
- Technical support on electrical and computer systems, installation monitoring, maintenance activities.

LANGUAGES

French: fluent · Bambara: native · English: technical